



TECHNICAL DOCUMENTATION

SenseMyPet

Privacy-first, on-device behavioural **screening** for dogs & cats — audio + camera signal processing, cautious evidence fusion, zero raw-media retention.

React 19 · Vite 8 PWA

AudioWorklet + Web Workers

On-device DSP

IndexedDB local history

No raw media leaves the device

Domain	sensemypet.com
Scope	Dogs + Cats only
Branch	feat/dog-cat-privacy-screening
Document	Installation · Architecture · UML · ERD · Privacy
Status	Build  · Typecheck  · Fixture harness 
Generated	2026-07-06

01 System Overview & What Changed

SenseMyPet screens observable stress/relaxation signals from a pet's sound and body language. It never claims a diagnosis. This rebuild made the app dogs-&-cats-only, honest about its evidence, and fully local.

DESIGN PRINCIPLES

- **Never invent evidence.** Degrade to `INSUFFICIENT` / `UNSUPPORTED` instead of forcing a verdict.
- **On-device only.** No raw audio/video, features, or embeddings leave the device.
- **Cautious language.** Screening indicators, not diagnoses.
- **Temporal evidence.** No classification from a single frame.
- **Measured, not claimed.** Accuracy reported only by the offline fixture harness.

FIVE CANONICAL OUTPUT CLASSES

`RELAXED` `POSSIBLE_STRESS` `POSSIBLE_ANXIETY`
`INSUFFICIENT_EVIDENCE` `UNSUPPORTED_SUBJECT`

Defined once in `src/lib/screening.ts` and shared by the Web Worker, engine, and UI.

Before → After

Area	Before (audited)	After (this build)
Species	Dog / Cat / Horse	Dog / Cat only — Horse removed end-to-end
Camera "AI"	Human MoveNet misapplied; silent <code>Math.sin</code> fabrication of posture + embedding	Honest gate: <code>subject: 'unvalidated'</code> , contributes no evidence; no CDN model
Audio subject gate	None — any tonal sound force-fit to nearest emotion	Species/context floor → non-pet audio ⇒ <code>UNSUPPORTED</code>
Reference embeddings	Live & reference builders had silently drifted (dims 35–511)	Single shared <code>assembleEmbedding()</code> — byte-for-byte parity
Persistence	Uploaded features + 512-d embedding to Supabase; UI claimed "nothing sent"	All user writes removed; history in <code>IndexedDB</code> ; Supabase read-only for reference fixtures
Language	"High Anxiety", "Anxiety Index", "Ready for production", "FP < 2%"	Cautious screening copy + mandatory disclaimer & verified privacy statement

Verification snapshot. `npm run build` ✓ `tsc -b` ✓ `npm test` ✓ — 7/7 negatives rejected, 0.0% false-positive rate, 100% taxonomy mapping. Network audit: the only calls in `src/` are two reference-fixture *downloads*.

Node 18+ and npm. Windows/macOS/Linux. The app runs fully offline; Supabase is optional and read-only.

1 · INSTALL & RUN

```
git clone <repo> && cd petsensor
npm install
npm run dev           # Vite dev server
npm run build         # tsc -b && vite build → dist/
npm run preview       # serve production build
npm run lint          # eslint
npm test              # offline screening fixture
harness
```

2 · ENVIRONMENT (OPTIONAL)

Create `.env` only if you want to download labelled reference fixtures. When unset, the app uses bundled signatures and makes **no network calls**.

```
VITE_SUPABASE_URL=https://<proj>.supabase.co
VITE_SUPABASE_ANON_KEY=<anon-read-key>
```

Bucket `pet-behavior-reference-audio` must be readable by the anon key (download-only).

Runtime requirements & permissions

Capability	Used for	Notes
Microphone (getUserMedia)	Audio screening	16 kHz mono; released on stop
Camera (getUserMedia)	Framing / future body-language	Optional; graceful "Audio Sensing Only" fallback
AudioWorklet	Low-latency framing	Falls back to ScriptProcessorNode
Web Workers	Off-thread DSP + frame handling	<code>audioWorker</code> , <code>videoWorker</code>
IndexedDB	Local scan history + reference cache	No server storage
Service Worker (PWA)	Offline install	vite-plugin-pwa, autoUpdate

Tech stack

UI

React 19, react-router 7, framer-motion, lucide-react, Tailwind-merge

BUILD

Vite 8 (Rolldown), TypeScript 6, vite-plugin-pwa, ESLint 10

SIGNAL

Web Audio API, AudioWorklet, Web Workers, custom FFT/MFCC DSP

03 System Architecture

A layered, on-device pipeline. The React UI drives one `UnifiedSensingEngine`, which runs audio DSP in a worker and consumes camera frames, then fuses independent evidence channels into a single cautious screening result.

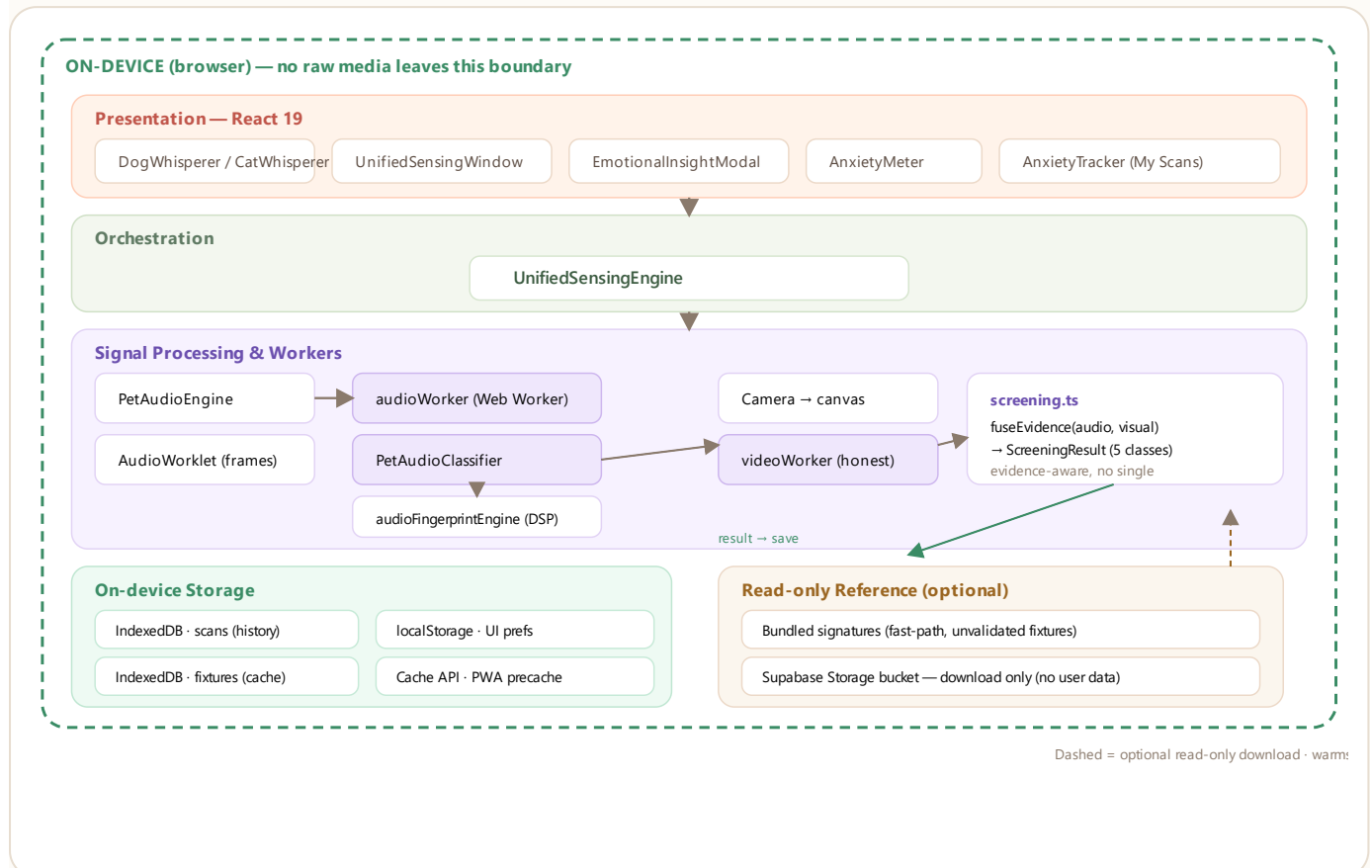


Figure 3.1 — Layered on-device architecture. The dashed boundary is the privacy perimeter; only reference fixtures may cross it (inbound, read-only).

Module responsibilities

Module	Responsibility
<code>lib/screening.ts</code>	5 output classes, display copy, <code>fuseEvidence()</code> , <code>similarityToConfidence()</code>
<code>lib/unifiedEngine.ts</code>	Orchestrates audio + camera, builds evidence channels, emits <code>UnifiedResult</code> , saves history
<code>lib/audioPipeline.ts</code>	<code>PetAudioEngine</code> : mic → AudioContext → AudioWorklet → worker; callbacks
<code>lib/audioWorker.ts</code>	Thin Web-Worker wrapper around the classifier
<code>lib/audioClassifier.ts</code>	<code>PetAudioClassifier</code> : pure decision core (shared with the test harness)
<code>lib/audioFingerprintEngine.ts</code>	FFT, mel/MFCC, spectral features, <code>assembleEmbedding()</code> , matching, temporal buffer, FP gate
<code>lib/petEmotionLibrary.ts</code>	Bundled unvalidated signatures + reference embeddings
<code>lib/videoWorker.ts</code>	Honest camera worker — releases frames, asserts nothing
<code>lib/localHistory.ts</code>	IndexedDB scan history (add/get/clear)
<code>lib/referenceLoader.ts</code>	Read-only fixture download → embed → cache (no-op when unconfigured)

Microphone PCM is framed by an AudioWorklet and classified off-thread. Reference and live features pass through the **same** code, so embeddings are directly comparable.

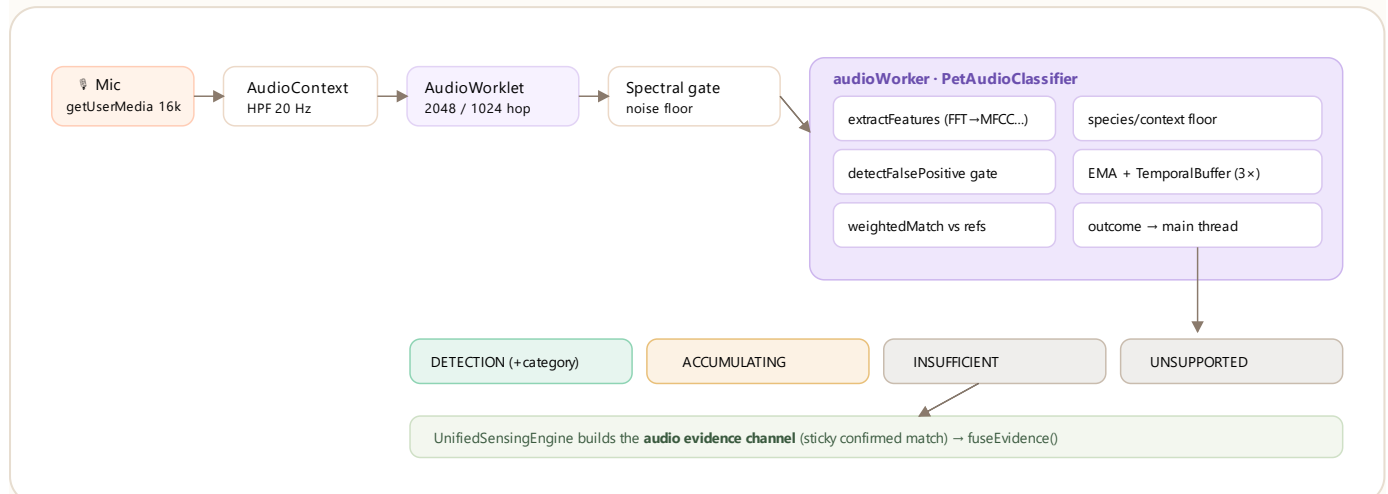


Figure 4.1 — Audio data flow. Raw PCM is transferred (zero-copy) to the worker; only a typed outcome returns. No waveform is retained or transmitted.

FEATURES (PER 2048-SAMPLE FRAME)

FFT (radix-2) → magnitude/power · RMS · ZCR · spectral centroid / rolloff / flux / flatness · 5 sub-band ratios · 26-mel → 13 MFCC · 12-bin chroma · F0 (autocorrelation) · top-5 spectral peaks · 512-d L2-normalized embedding.

GATES & THRESHOLDS

Silence RMS < 0.015 → INSUFFICIENT

Speech / noise / click → UNSUPPORTED

Species floor bestRaw < 0.35 → UNSUPPORTED

Confidence gate cosine ≥ 0.92

Temporal 3 consecutive · var < 15%

Parity guarantee. Live and reference embeddings are produced by one exported `assembleEmbedding()`. A prior silent drift in dims 35–511 was eliminated; the offline harness asserts taxonomy mapping stays intact.

05 Screening Decision & Evidence Fusion

Two independent channels (audio, visual) feed `fuseEvidence()`. Order matters and no single weak signal can dominate. Today the visual channel is honestly "insufficient" until a real pet-vision model ships.

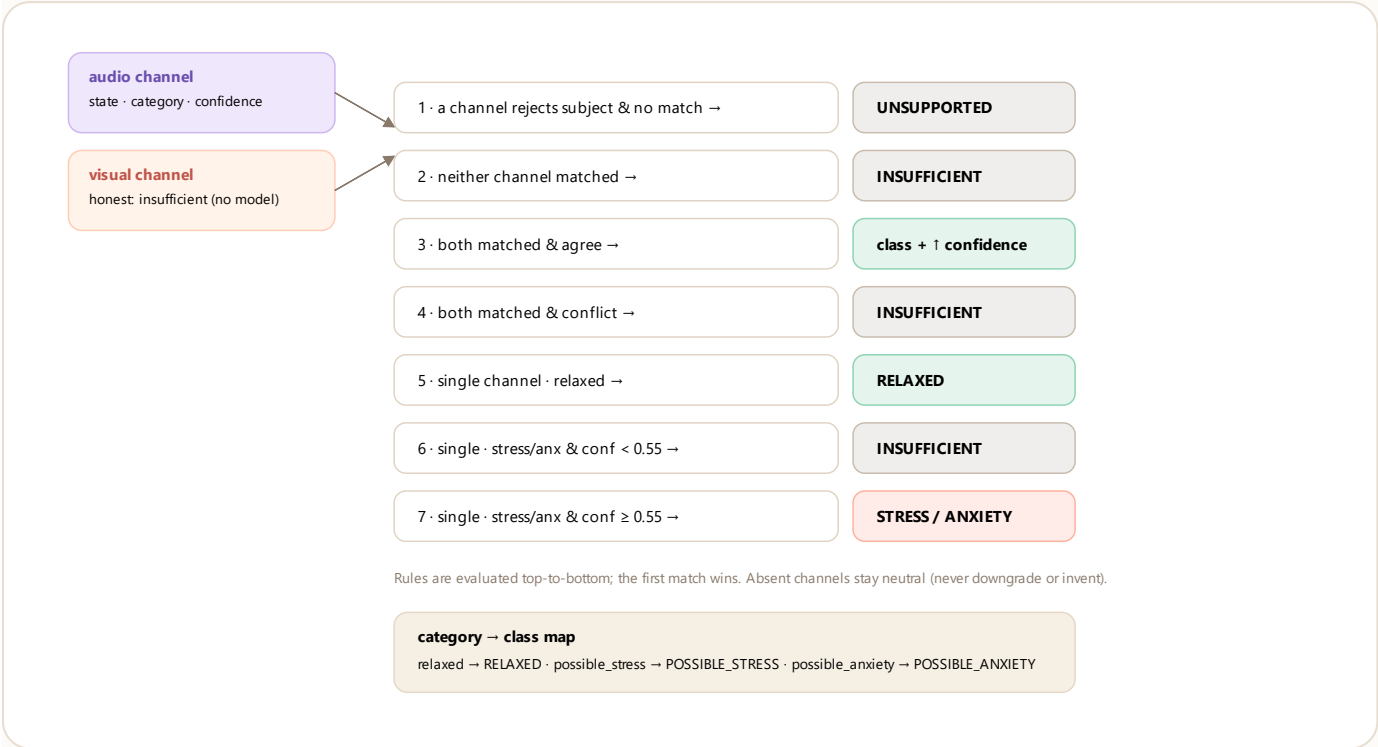


Figure 5.1 — `fuseEvidence()` decision ladder. Confidence is a compressed band (0.45–0.90) mapped from the ≥ 0.92 cosine gate, so the UI never reads as certainty.

Signature → screening category mapping

Species	Signature	Category	Screening class
Dog	Calm & Resting / Playful	relaxed	RELAXED
Dog	Growling / Distress Whimper	possible_stress	POSSIBLE_STRESS
Dog	Anxious Barking / Repeated Whining	possible_anxiety	POSSIBLE_ANXIETY
Cat	Calm Purring	relaxed	RELAXED
Cat	Repeated Distress Meowing / Prolonged Cry	possible_stress	POSSIBLE_STRESS
Cat	Hissing	possible_anxiety	POSSIBLE_ANXIETY

Key classes, interfaces, and their relationships. The pure `PetAudioClassifier` is shared by the Web Worker and the offline test harness.

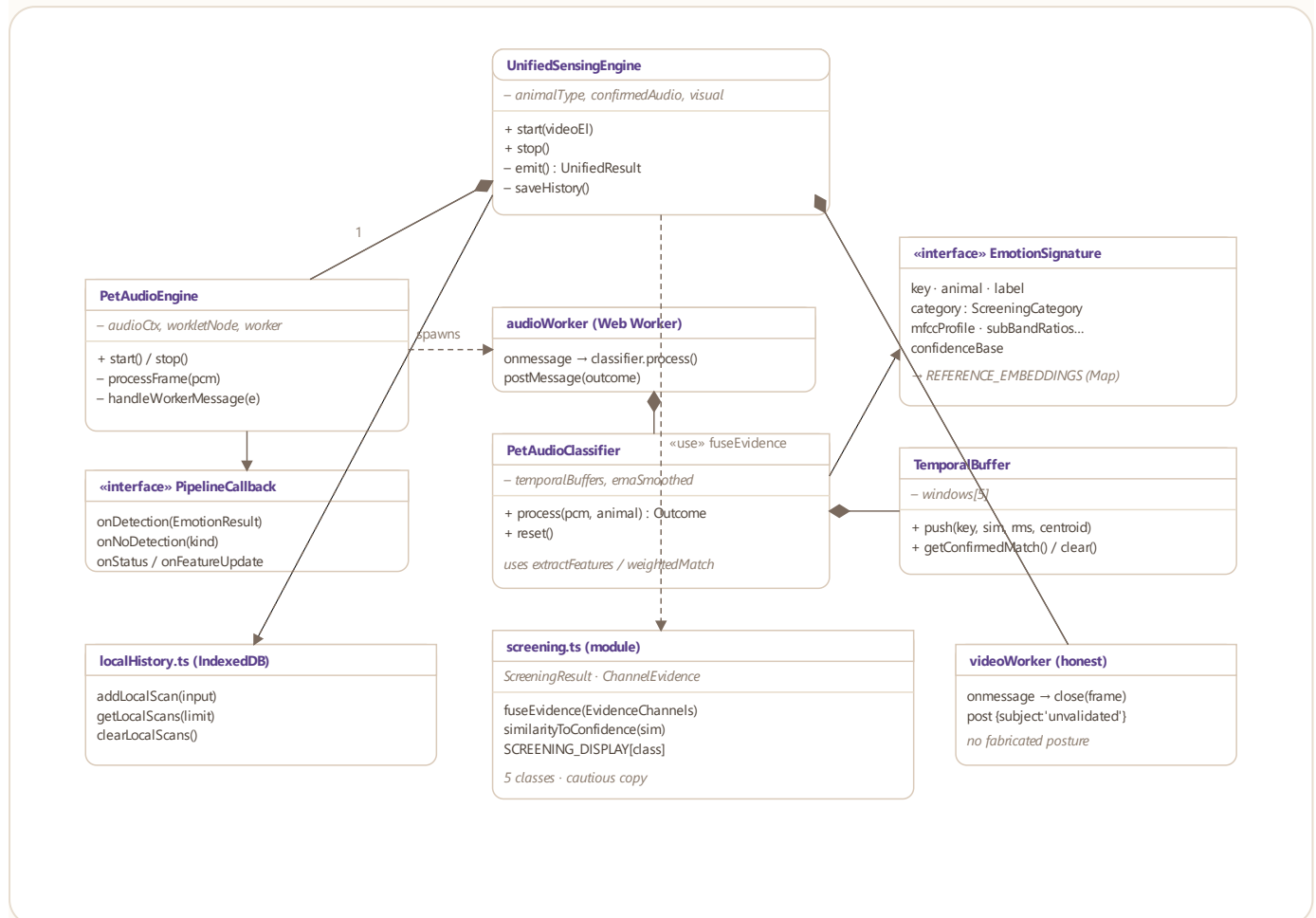


Figure 6.1 — Class diagram. ◆ = aggregation/composition · dashed ▷ = «use»/dependency. The classifier is deliberately pure so the harness exercises the exact production path.

Lifecycle of a single scan, from tapping "Start" to a saved on-device summary.

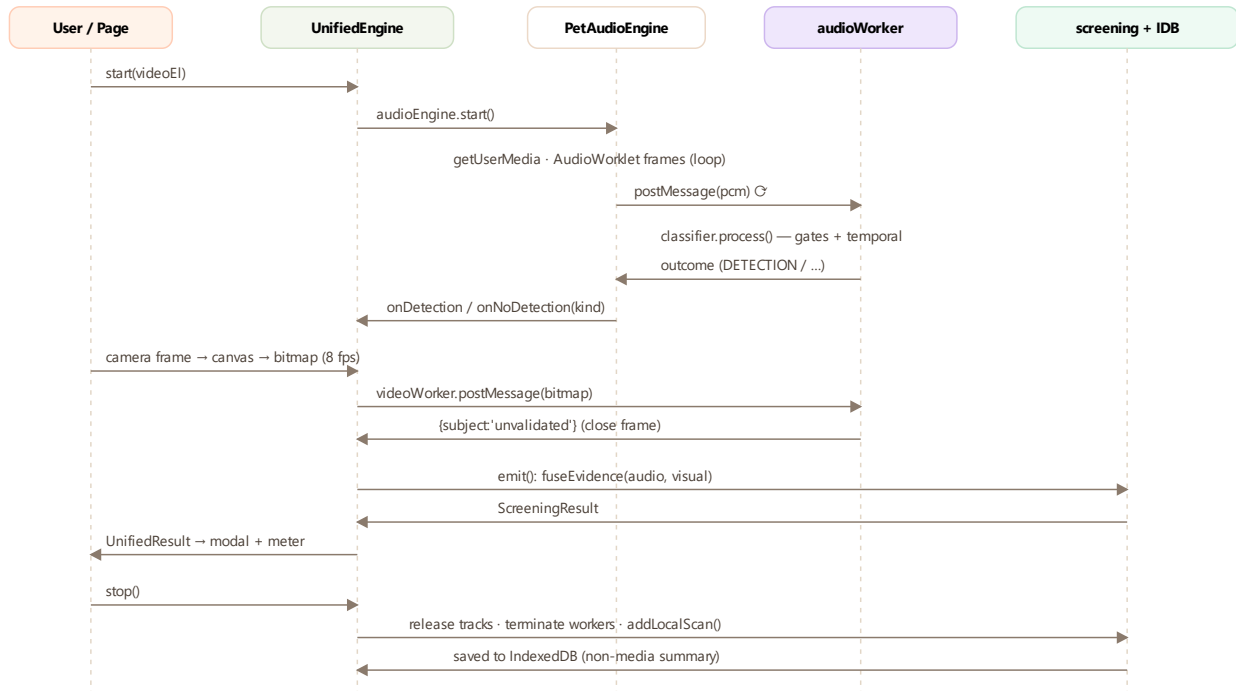
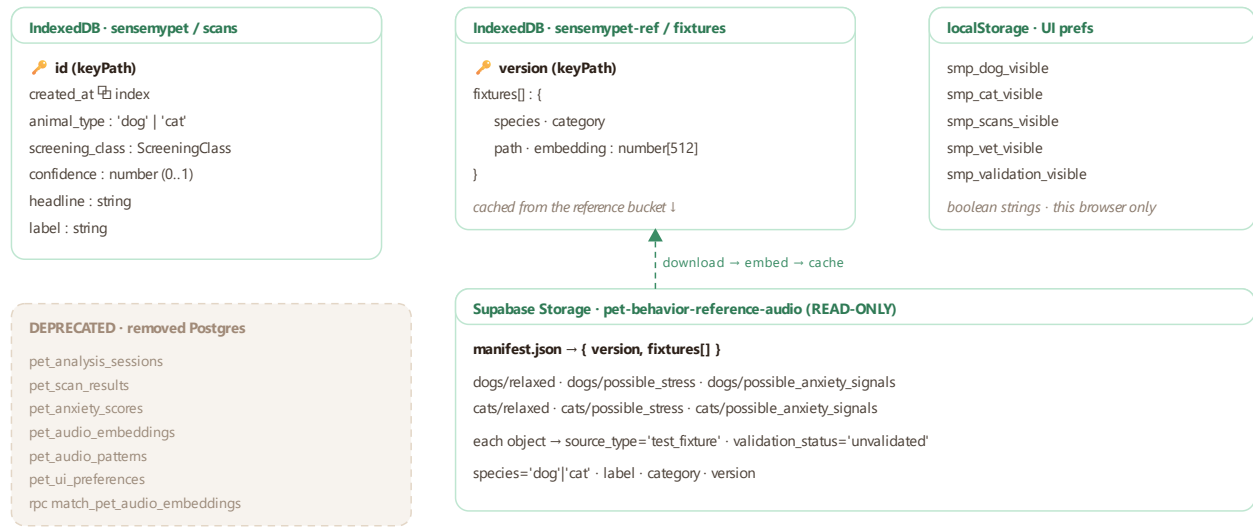


Figure 7.1 — Scan sequence. On **stop()** the engine releases all *MediaStream* tracks, terminates workers, closes bitmaps, and persists only a non-media summary.

The app is local-first. The "database" is on-device (IndexedDB + localStorage). The only server-side structure is a read-only Supabase Storage bucket of reference fixtures. Legacy Postgres tables are retained here as **deprecated** history.



All user-data writes/reads to these tables were removed. No pgvector search on user embeddings; no session/result rows leave the device.

Figure 8.1 — Entity model. Solid teal = active on-device stores; amber = read-only server reference; dashed grey = removed legacy schema (shown for provenance only).

Why "ERD-lite"? With all relational user data removed, there are no cross-entity foreign keys among user records — history is a flat, non-reconstructable summary store. This is intentional: it makes the privacy claim verifiable.

The perimeter is the device. Raw audio/video never crosses it; derived features/embeddings never cross it. Only reference fixtures may enter (inbound, read-only).

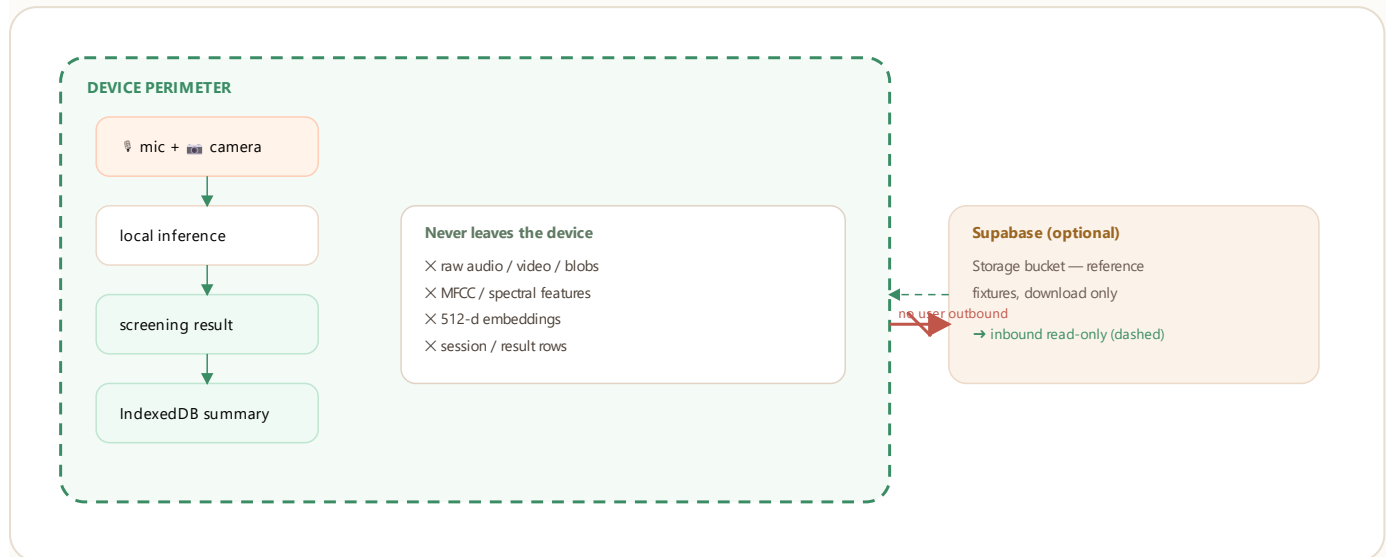


Figure 9.1 — Privacy boundary. Static audit: the only network calls in `src/` are two `storage.download()` reference fetches. There is no `.upload()`, `.insert()`, `.rpc()`, `fetch()`, or base64/blob transmission of user data.

User-facing statement (verified true): "Audio and camera input are processed on your device for the active scan. Raw audio and video are not stored on our servers."

Lifecycle on stop: release `MediaStream` tracks · clear buffers · close `ImageBitmaps` · terminate workers · null `srcObject` · close `AudioContext`.

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Test Matrix & Measured Results

The offline harness (`npm test`) feeds synthetic PCM through the exact live decision core and prints a confusion matrix. Numbers are **measured** against synthetic, unvalidated fixtures — not a real-world accuracy claim.

Expected behaviour

Input	Kind	Expected	Result
silence	negative	INSUFFICIENT / UNSUPPORTED	✓ INSUFFICIENT
white-noise / fan	negative	not a pet state	✓ UNSUPPORTED
traffic / rumble	negative	not a pet state	✓ UNSUPPORTED
human speech	negative	not a pet state	✓ UNSUPPORTED
music	negative	not a pet state	✓ UNSUPPORTED
appliance hum	negative	not a pet state	✓ INSUFFICIENT
single click	negative	not a pet state	✓ INSUFFICIENT
dog / cat synthetic	positive	correct category	□ recall 0/2 (synthetic — honest)

NEGATIVES REJECTED

7 / 7

100% — no negative became a pet state

FALSE-POSITIVE RATE

0.0%

on negatives

TAXONOMY MAPPING

100%

10/10 category→class

Honesty note. Synthetic positives are not tuned to clear the 0.92 confirmation gate, so measured recall on them is low (0/2) — this is expected and documented. Real recall requires real labelled fixtures, which is exactly what the reference-fixture pipeline is built to ingest. No ">95%" figure is asserted anywhere.

Confusion matrix (raw-PCM synthetic)

group	RELAXED	STRESS	ANXIETY	INSUFFICIENT	UNSUPPORTED
negative	0	0	0	3	4
positive	0	0	0	1	1

11 Module Map & Change Summary

New / rewritten modules

Path	Kind	Purpose
lib/screening.ts	new	Taxonomy + fusion
lib/audioClassifier.ts	new	Pure decision core
lib/localHistory.ts	new	IndexedDB history
lib/referenceLoader.ts	new	Read-only fixture download/cache
components/ScreeningDisclaimer.tsx	new	Disclaimer + privacy statement
lib/__tests__/screeningHarness.ts	new	Offline confusion-matrix harness
lib/unifiedEngine.ts	rewritten	Fusion + local-only persistence
lib/videoWorker.ts	rewritten	Honest camera gate
lib/supabase.ts	rewritten	Read-only reference client
components/EmotionalInsightModal.tsx · AnxietyMeter.tsx	rewritten	Screening-driven UI
pages/HorseWhisperer.tsx · components/CameraView.tsx · public/manifest.json	deleted	Scope / dead code / redundant

Appendix A — Output class copy

Class	Headline
RELAXED	Relaxed indicators observed
POSSIBLE_STRESS	Possible stress-related signals observed
POSSIBLE_ANXIETY	Possible anxiety-related signals observed
INSUFFICIENT_EVIDENCE	Insufficient evidence — try again in a quieter setting
UNSUPPORTED_SUBJECT	Unsupported subject — this feature supports dogs and cats only

Appendix B — Commands

Command	What it does
npm run dev	Vite dev server (HMR)
npm run build	tsc -b then vite build → dist/ (+ service worker)
npm test	Offline screening fixture harness → confusion matrix + metrics
npm run lint	ESLint (pre-existing debt not gated by build)
npm run preview	Serve the production build locally

Not a diagnosis. SenseMyPet provides AI-assisted behavioural screening based on observable sound and visual patterns. Results are informational only and are not a veterinary diagnosis. Behavioural signs may have many causes. If you are concerned

about your pet's health or behaviour, consult a qualified veterinarian.

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